Automatic Interpretation of Ancient Egyptian Texts for Education and Research

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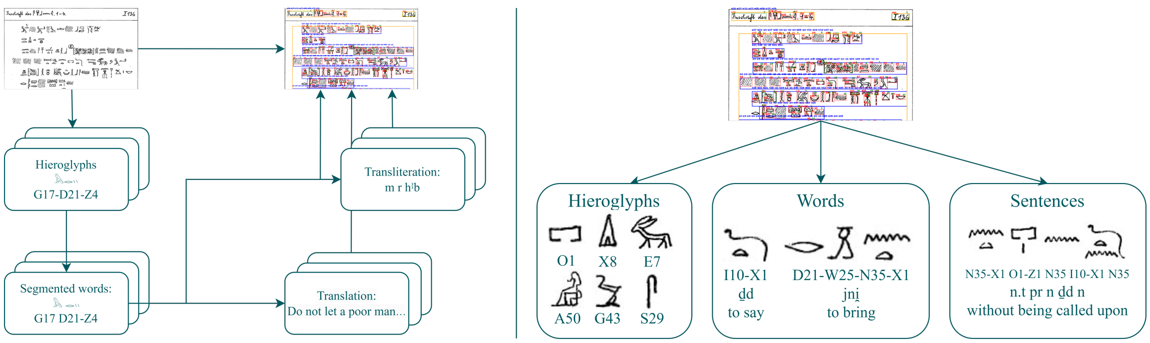

Fig. 1. **Left**: System architecture showing the flow from input image to final text interpretation. The pipeline handles recognition, word segmentation, transliteration, and translation. **Right**: Example outputs at three levels — glyphs, words, and sentences — with Gardiner codes, transliterations, and translations.

We introduce a pipeline for interpreting Ancient Egyptian hieroglyphic texts that combines OCR, transliteration, and translation. Designed for the low-resource nature of available data, our system improves accessibility for learners and efficiency for researchers. We evaluate its performance on a newly collected, diverse dataset reflective of real-world conditions.

CCS Concepts: • **Applied computing** → **Interactive learning environments**.

Additional Key Words and Phrases: ocr, translation, education

ACM Reference Format:

Maksim Golyadkin, Innokentiy Humonen, Yanis Plevokas, Ekaterina Bureeva, Ekaterina Alexandrova, and Ilya Makarov. 2018. Automatic Interpretation of Ancient Egyptian Texts for Education and Research. In *Proceedings of Make sure to enter the correct conference title from your rights confirmation email (Conference acronym 'XX).* ACM, New York, NY, USA, 4 pages. https://doi.org/XXXXXXX.XXXXXXX

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1 Introduction

Ancient Egyptian is a complex and visually rich language that offers deep insights into the civilization's culture, religion, and daily life. Whether for students or experts, interpreting Egyptian texts often involves time-consuming searches through reference materials, with even a single unknown glyph potentially slowing down the entire process. These challenges make tools that support automatic recognition and interpretation of hieroglyphic texts highly valuable for both education and research.

Currently, there are no publicly available open-source systems capable of Ancient Egyptian text interpretation. Existing research [4] typically focuses on narrow and uniform datasets. This overlooks the wide variety of real-world sources, including handwritten texts, stylized drawings, and photographs of physical artifacts. Such diversity introduces significant variability in appearance and structure, posing major challenges for generalization. We are the first to present a system designed to handle this broad spectrum of input types, and we will make it openly available to support further research and educational use.

In this work, we present a pipeline (see Figure 1) for Ancient Egyptian text interpretation that integrates optical character recognition, transliteration, and translation, trained on a newly collected and diverse dataset. The system is specifically optimized for the current state of available data, offering strong generalizability in the challenging low-resource setting of hieroglyphic texts. We further provide an evaluation to assess the performance of existing methods under current data conditions.

2 Approach

Our pipeline consists of four sequential components: an OCR module that identifies hieroglyphs and predicts Gardiner codes, a segmentation block that groups these codes into words, and two sequence-to-sequence models for transliteration and translation. Each module is trained independently and integrated into a unified system, enabling end-to-end interpretation. This modular design allows for task-specific optimization, improving quality under data-scarce conditions.

Training data and labels are sourced from the Thesaurus Linguae Aegyptiae (TLA) [3], which provides digitized hieroglyphic texts with aligned transliterations and translations across diverse Ancient Egyptian sources.

The output of our pipeline is structured across three levels of interpretation: individual hieroglyphs, segmented words, and full sentences. At the first level, each hieroglyph is labeled with its Gardiner code. At the second level, words are represented as sequences of Gardiner codes, with transliterations and translations retrieved from a dictionary. At the sentence level, each sentence is composed as a sequence of words in Gardiner code, accompanied by transliteration and translation.

2.1 OCR

Our OCR module follows a three-stage pipeline, with each model handling a specific subtask to improve robustness under limited data. It begins with region segmentation using YOLOv11, which detects rows and columns of hieroglyphic text. A second model then localizes individual hieroglyphs. These symbols are passed to a classifier that predicts both the Gardiner code and orientation.

The predicted orientations are used in a topological sorting algorithm to reconstruct the correct reading order. For hieroglyph detection and classification, we fine-tuned several architectures and selected the best-performing models based on test set accuracy (see Table 1).

To train and evaluate our OCR pipeline, we constructed a dataset using digitized texts available in the TLA. We found corresponding image sources, such as scans or photographs, and manually aligned them with the TLA annotations. The final dataset includes 28 fully annotated images representing diverse writing styles. From these pages, we also derived a classification dataset comprising 12,750 labeled hieroglyph crops.

Hieroglyph Detection		**Hieroglyph Classification**	
Model	**mAP**	**Model**	**F1**
YOLO11s	0.58	ResNet-50	0.65
YOLO11x	0.71	Swin-B	0.84
DINO (ResNet-50)	0.43	Swin-L	0.89
DINO (Swin-L)	0.69	**ConvNeXt-B**	**0.91**
CO-DETR (Swin-L)	**0.73**	ConvNeXt-L	0.85

Table 1. **Performance comparison for detection and classification tasks.** Left: object detection models evaluated on hieroglyph detection using mAP@[0.5:0.95]. Right: classification models evaluated using F1-micro.

2.2 Word Segmentation and Transliteration

Word segmentation is a task of dividing continuous text into words, crucial for OCR outputs and ancient scripts. Middle Egyptian poses particular challenges due to its logographic-phonographic system and lack of spacing.

We augmented the TLA corpus with word-boundary labels (90-5-5 train-val-test split) and evaluated three approaches: BERT for sequence labeling MaxMatch (MM) (longest dictionary match), and Greedy Unigram (GU) (highest-frequency segmentation). MM achieved the best F-score 0.81, outperfoming BERT, GU and separator after each synbol baseline (0.73, 0.72 and 0.49, respectively)

2.3 Transliteration

We implemented a baseline transliteration model using a random forest classifier with bag-of-hieroglyphs word representations. This simple approach maps each word to its most frequent transliteration, achieving 85% accuracy and demonstrating that there are dominant transliterations for common words. Although effective, this method leaves significant room for improvement.

2.4 Translation

The final translation dataset combines two resources: *tla-Earlier_Egyptian_original-v18-premium* ,containing hieroglyphic script, transliteration, linguistic annotation, and German translations and *phiwi/bbaw_egyptian*, 35,500 sentence alignments with hieroglyphic encodings, transcriptions, and English/German translations.

Model	from transcription		from hieroglyphs	
	RougeL	BLEU	BLEU	RougeL
byT5	0.5329	27.33	27.33	0.5329
m2m-100*	0.3862	14.06	-	-
NLLB w/o DPO	**0.5996**	**35.91**	25.57	0.4320
NLLB with DPO	0.5845	28.82	**33.57**	**0.5334**

Table 2. Evaluation results for translation tasks. Best scores for each task are marked in bold. Note [1] reports 36.4 BLEU for hieroglyphical task, unverifiable with our data.

Considering Middle Egyptian's low-resource status, we fine-tuned *facebook/nllb-200-distilled-600M* [2] No Language Left Behind model. ByT5, a token-free version of the T5 model, was used as a baseline. We also additionally fine-tuned NLLB using Direct Preference Optimization (DPO) loss function: model-generated translations as negative outputs and dataset translations as preferred ones.

Table 2 shows the DPO-finetuned NLLB achieving the best performance of 33.57 BLEU for the hieroglyphic translation.

Human evaluation involved a pairwise arena-like comparison and direct assessment. Pairwise: 92% annotators agreement (4 annotators, 50 texts). Results: 17-8-21 (byT5 vs. NLLB w/o DPO) and 4-10-6 (NLLB w/o DPO vs. NLLB with DPO) Direct: 5-point rating with 3 criteria Sense, Coherence and Style; the best NLLB with DPO model shows 4.34, 4.6 and 4.6 scores and 0.32 named entity errors rate.

3 Case Study

To evaluate the effect of our approach on text interpretation speed for students and non-experienced readers, we conducted a user study with five participants: three Egyptology students and two non-experts. Each interpreted three lines of hieroglyphic text manually and three using our approach. Egyptologists performed hieroglyph mapping, transliteration, and translation, while non-experts focused on mapping only.

Task	**Manual**	**Our Approach**	**Boost**
Egy-Mapping	35	6	5.8×
Egy-Translit.	23	15	1.5×
Egy-Transl.	20	12	1.7×
NExp-Mapping	95	34	2.8×

Table 3. **Model performance with and without refinement.** CER is reported for OCR; BLEU is reported for transliteration and translation.

Table 3 shows that our approach substantially accelerates the mapping stage, thanks to its accurate suggestions that reduce manual lookups. Participants noted that even rare signs were often predicted correctly, saving considerable time and effort. Although the gains in transliteration and translation were more modest, Egyptologists reported that pipeline outputs helped them grasp overall meaning more quickly.

4 Conclusion and Future Work

We presented a complete pipeline for the interpretation of Ancient Egyptian hieroglyphic texts, integrating OCR, transliteration, and translation. Our system demonstrates promising results in both automated evaluation and real-user studies. Future work includes integrating the pipeline into an interactive tool and improving model performance.

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